Physical Chemistry (I)

Lecture 23

In the Last Class

- Activities of ions
  - Debye-Hückel Limiting Law
  - Activity and cell potential

- Maxwell-Boltzmann statistics
  - Boltzmann factor and partition function
  - Classical statistical mechanics: phase space

Mean Activity Coefficient

For a salt M_pX_q (= $pM^+ + qX^-$),

the mean activity coefficient is defined as

$$\gamma_\pm = \left(\gamma_+^p \gamma_-^q\right)^{1/(p+q)}$$

so that

$$\mu_J = \mu_J^{\mathrm{ideal}} + RT \ln \gamma_\pm$$

Debye-Hückel Limiting Law

From the work of charging an ion,

$$\log_{10}\gamma_{\pm} = -A|z_{+}z_{-}|I^{1/2}$$

where $A = 0.509$ for an aqueous solution at 25°C

Review

Ion Activity and Cell Potential

For the Harned cell:

$$\mathrm{Pt(s)\,|\,H_2(g)\,|\,HCl}(aq,b)\,|\,\mathrm{AgCl(s)\,|\,Ag(s)}$$

$$(1/2)\mathrm{H_2(g)} + \mathrm{AgCl(s)} \longrightarrow \mathrm{HCl}(aq) + \mathrm{Ag(s)}$$

Combining the Nernst equation and the Debye-Hückel law,

$$\underbrace{E_{\mathrm{cell}} + \frac{2RT}{F}\ln\frac{b}{b^{\ominus}}}_{y} = E^{\ominus} + \underbrace{\frac{2ART\ln 10}{F}}_{a}\underbrace{\sqrt{\frac{b}{b^{\ominus}}}}_{x}$$

Boltzmann Factor

In general, for a microstate s with energy E,

$$P(s) \propto e^{-E(s)/k_\mathrm{B}T} \text{ (Boltzmann factor)}$$

or,

$$P(s) = \frac{1}{q} e^{-E(s)/k_\mathrm{B}T}$$

where the partition function q is

$$q = \sum_{s} e^{-E(s)/k_\mathrm{B}T}$$

Phase Space

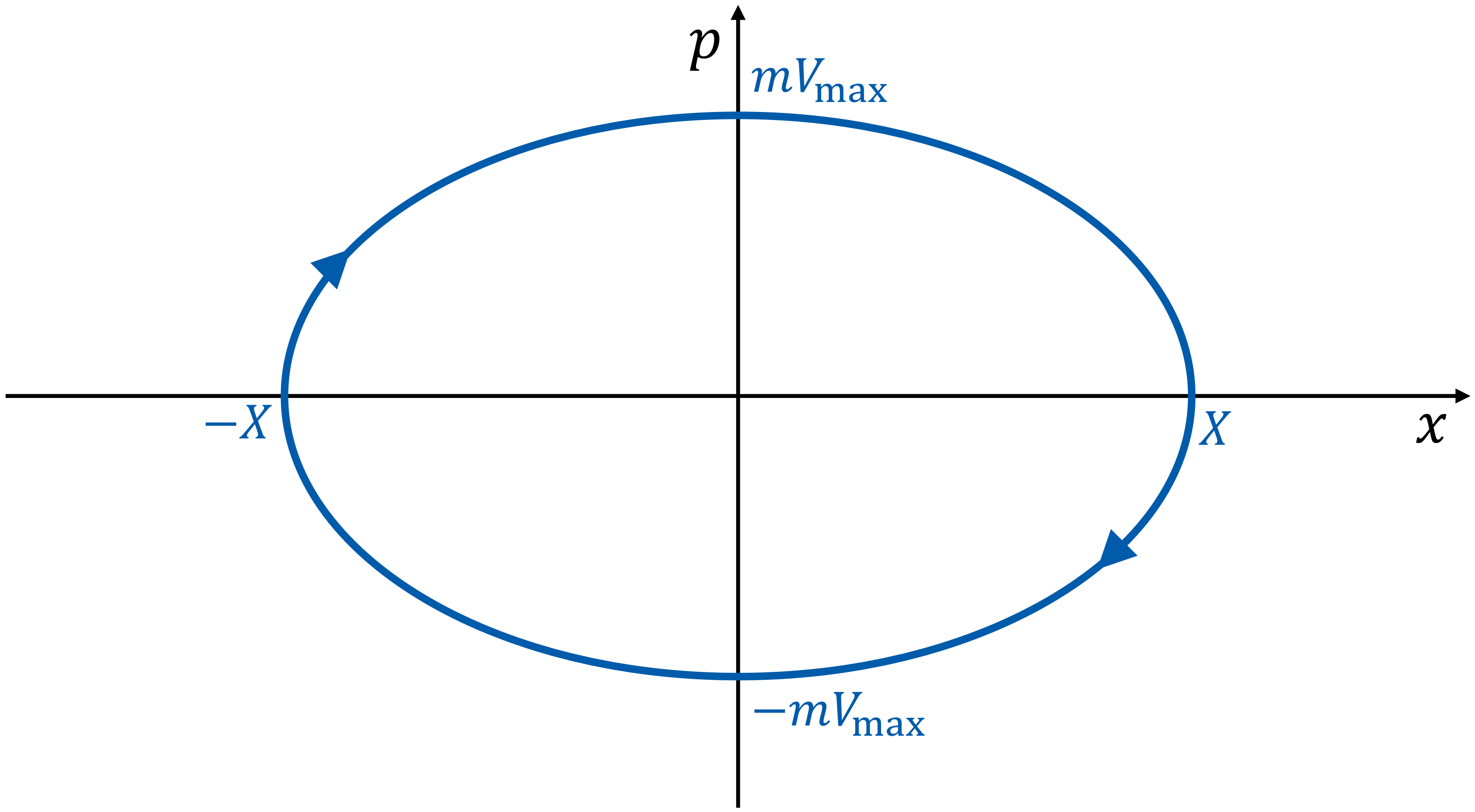
Space of all possible microstates of a system

- Classical mechanics:

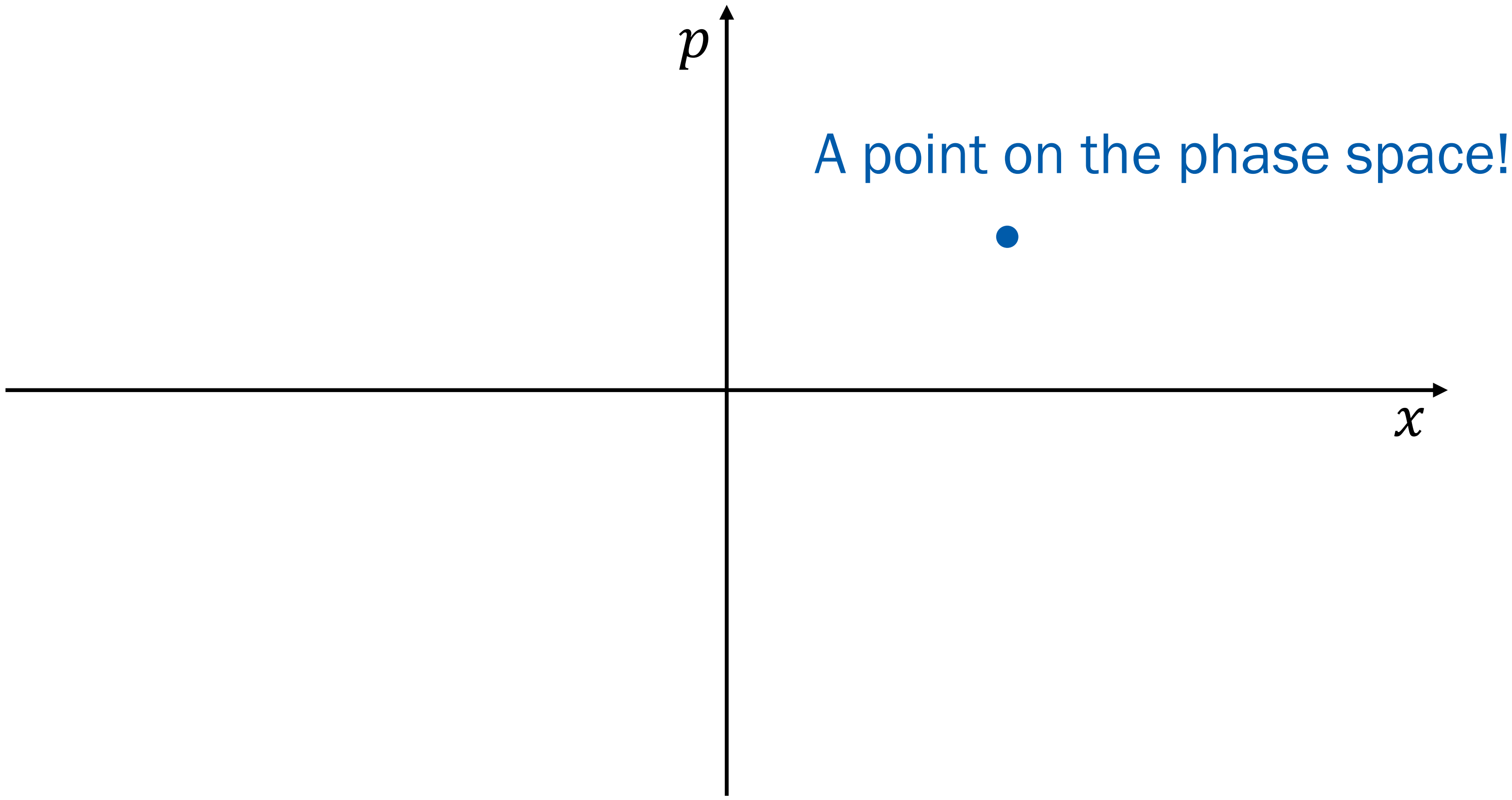
Space of all possible values of

position and momentum variables

1-Dimensional Spring



Microstate of Classical Particle



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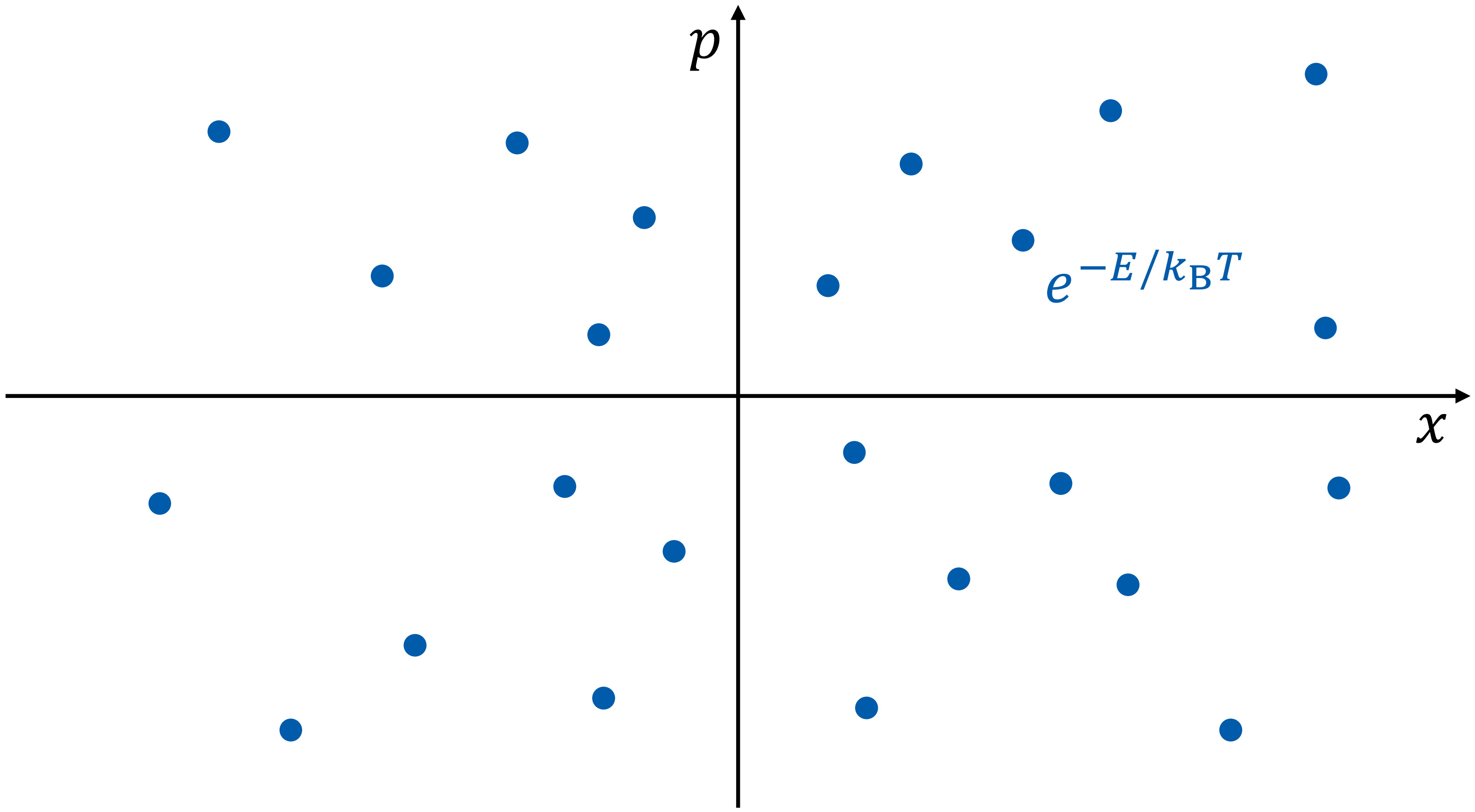
Partition Function



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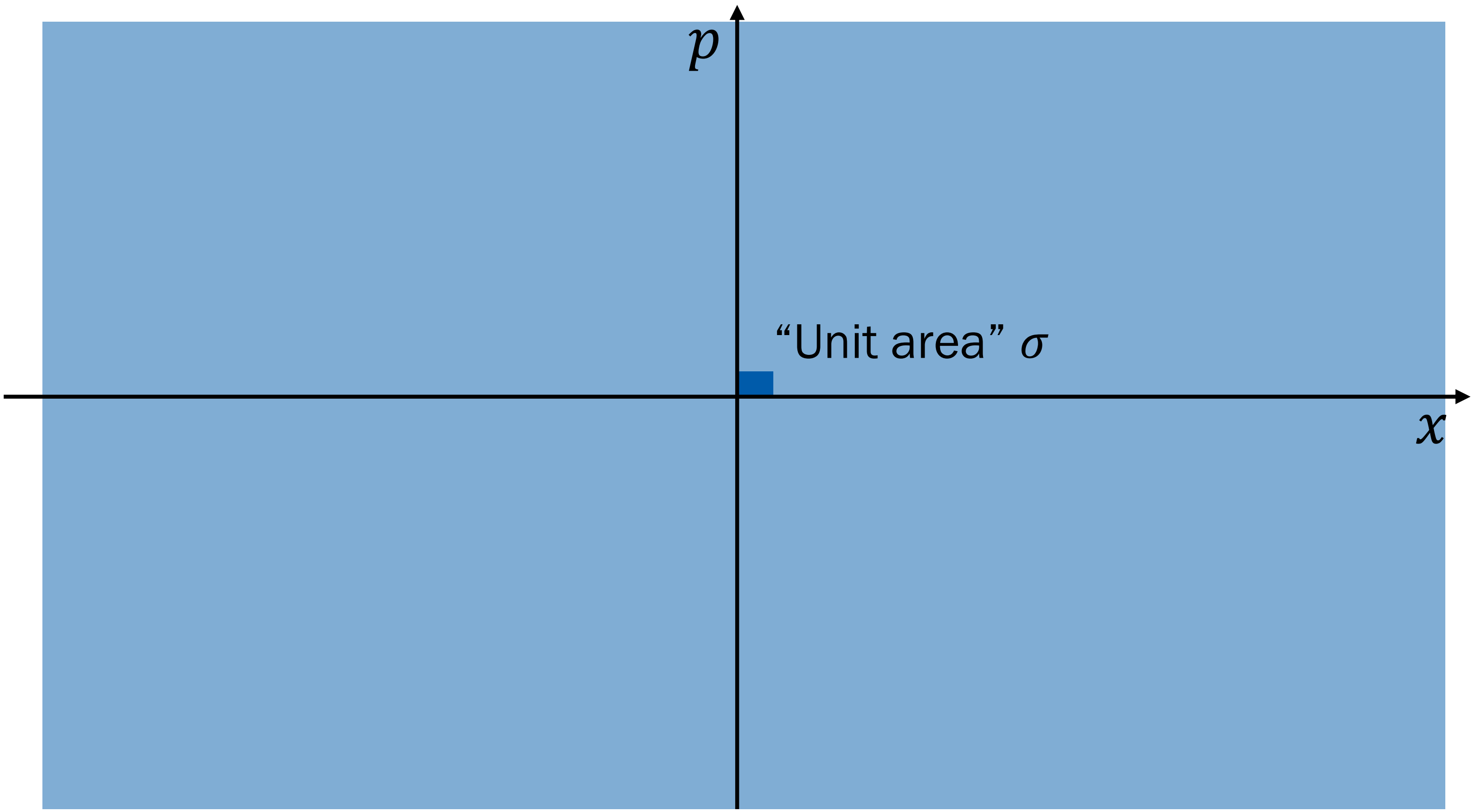
Partition Function



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Partition Function

$$q = \sum_{s} e^{-E(s)/k_{\mathrm{B}}T} \rightarrow q = \frac{1}{\sigma}\int\int e^{-E(x,p)/k_{\mathrm{B}}T}\, dx dp$$

for a 1-dimensional single particle

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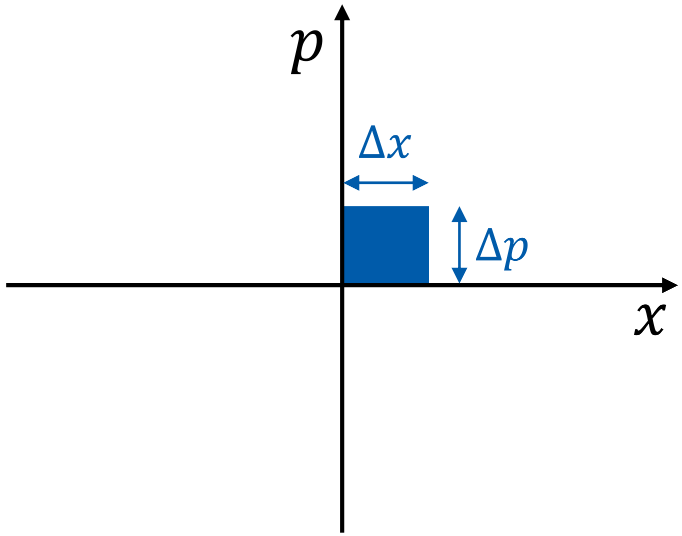
What is σ?

From Heisenberg's uncertainty principle,

$\Delta x \Delta p \geq h$ (Planck's constant)



Take $\sigma = h$!

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Partition Function

$$q = \frac{1}{h} \int \int e^{-E(x,p)/k_{\mathrm{B}}T}\, dxdp$$

- Example: 1-dimensional spring

$$E(x,p) = \frac{1}{2}kx^2 + \frac{1}{2}mv^2 = \frac{1}{2}kx^2 + \frac{p^2}{2m}$$

$$q = \frac{1}{h} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-\frac{1}{k_{\mathrm{B}}T}\left(\frac{1}{2}kx^2 + \frac{p^2}{2m}\right)}\, dxdp$$

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1-Dimensional Spring

$$q = \frac{1}{h}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-\frac{1}{k_{\mathrm{B}}T}\left(\frac{1}{2}kx^2+\frac{p^2}{2m}\right)}\,dxdp$$

$$q = \frac{1}{h}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-\frac{kx^2}{2k_{\mathrm{B}}T}-\frac{p^2}{2mk_{\mathrm{B}}T}}\,dxdp$$

$$q = \frac{1}{h}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-\frac{kx^2}{2k_{\mathrm{B}}T}} \times e^{-\frac{p^2}{2mk_{\mathrm{B}}T}}\,dxdp$$

$$q = \frac{1}{h}\left[\int_{-\infty}^{\infty} e^{-\frac{kx^2}{2k_{\mathrm{B}}T}}\,dx\right]\left[\int_{-\infty}^{\infty} e^{-\frac{p^2}{2mk_{\mathrm{B}}T}}\,dp\right]$$

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Gaussian Integrals

$$q = \frac{1}{h}\left[\int_{-\infty}^{\infty} e^{-\frac{kx^2}{2k_\mathrm{B}T}}\,dx\right]\left[\int_{-\infty}^{\infty} e^{-\frac{p^2}{2mk_\mathrm{B}T}}\,dp\right]$$

Using $\int_{-\infty}^{\infty} e^{-\alpha x^2}\,dx = \sqrt{\pi/\alpha}$,

$$q = \frac{1}{h}\sqrt{\frac{2\pi k_\mathrm{B}T}{k}}\sqrt{2m\pi k_\mathrm{B}T} = \frac{2\pi k_\mathrm{B}T}{h}\sqrt{\frac{m}{k}}$$

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Into Higher Dimensions

- Single 1-dimensional particle

$$q = \frac{1}{h}\int\int e^{-E(x,p)/k_\mathrm{B}T}\,dxdp$$

- Single 3-dimensional particle

$$q = \frac{1}{h^3}\iiint\iiint e^{-E\left(x,y,z,p_x,p_y,p_z\right)/k_\mathrm{B}T}\,dxdydzdp_xdp_ydp_z$$

$$\rightarrow\ q = \frac{1}{h^3}\int\int e^{-E(\vec{r},\vec{p})/k_\mathrm{B}T}\,d\vec{r}d\vec{p}$$

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Higher and Higher

- Single 3-dimensional particle

$$q = \frac{1}{h^3} \int \int e^{-E(\vec{r},\vec{p})/k_\mathrm{B}T}\, d\vec{r} d\vec{p}$$

- N 3-dimensional (independent) particles

$$Q = q_1 \times q_2 \times q_3 \times q_4 \times \cdots \times q_N$$

$$Q = \frac{1}{h^{3N}} \left[\int \int e^{-E(\vec{r},\vec{p})/k_\mathrm{B}T}\, d\vec{r} d\vec{p} \right]^N$$

Indistinguishability

- N 3-dimensional distinguishable particles:

$$Q = \frac{1}{h^{3N}} \left[\int \int e^{-E(\vec{r},\vec{p})/k_\mathrm{B}T} \, d\vec{r} d\vec{p} \right]^N$$

- N 3-dimensional indistinguishable particles:

$$Q = \frac{1}{N! \, h^{3N}} \left[\int \int e^{-E(\vec{r},\vec{p})/k_\mathrm{B}T} \, d\vec{r} d\vec{p} \right]^N$$

Probability Distributions

	Discrete	Continuous
Variable	$k = 1,2,\dots,N$	x: real in $[x_1, x_2]$
Probability	$p(k)$	probability density $p(x)dx$
Normalization	$\sum_{k=1}^{N} p(k) = 1$	$\int_{x_1}^{x_2} p(x)\,dx = 1$
Average	$\langle k \rangle = \sum_{k=1}^{N} kp(k)$	$\langle x \rangle = \int_{x_1}^{x_2} xp(x)\,dx$
Expectation	$\langle f \rangle = \sum_{k=1}^{N} f(k)p(k)$	$\langle f \rangle = \int_{x_1}^{x_2} f(x)p(x)\,dx$

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Classical Statistical Mechanics

For a single particle,

- Variable: $\vec{r}$ and $\vec{p}$
- Probability:

$$\frac{1}{h^3}P(\vec{r},\vec{p})d\vec{r}d\vec{p} = \frac{1}{h^3}\frac{e^{-E(\vec{r},\vec{p})/k_\mathrm{B}T}}{q}d\vec{r}d\vec{p}$$

- Normalization:

$$\int\int\frac{1}{h^3 q}e^{-E(\vec{r},\vec{p})/k_\mathrm{B}T}\,d\vec{r}d\vec{p} = \frac{1}{q}\times q = 1$$

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Classical Statistical Mechanics

For a single particle,

- Probability:

$$\frac{1}{h^3} P(\vec{r}, \vec{p}) d\vec{r} d\vec{p} = \frac{1}{h^3 q} e^{-E(\vec{r},\vec{p})/k_{\mathrm{B}}T} d\vec{r} d\vec{p}$$

- Expectation value:

$$\langle f \rangle = \frac{1}{h^3} \int \int f(\vec{r}, \vec{p}) P(\vec{r}, \vec{p})\, d\vec{r} d\vec{p}$$

$$= \frac{1}{h^3 q} \int \int f(\vec{r}, \vec{p}) e^{-E(\vec{r},\vec{p})/k_{\mathrm{B}}T}\, d\vec{r} d\vec{p}$$

Internal Energy

- Internal energy is the expectation value of energy

$$U = \langle E \rangle = \frac{1}{h^3 q} \int \int E(\vec{r},\vec{p}) e^{-E(\vec{r},\vec{p})/k_\mathrm{B}T}\, d\vec{r} d\vec{p}$$

Define a new variable: $\beta = 1/k_\mathrm{B}T$

$$U = \frac{1}{h^3 q} \int \int E(\vec{r},\vec{p}) e^{-\beta E(\vec{r},\vec{p})}\, d\vec{r} d\vec{p}$$

Small Trick

$$U = \frac{1}{h^3 q} \int \int E(\vec{r}, \vec{p}) e^{-\beta E(\vec{r},\vec{p})} \, d\vec{r} d\vec{p}$$

Since $\partial e^{-ax} / \partial x = -a e^{-ax}$,

$$U = -\frac{1}{h^3 q} \int \int \left(\frac{\partial e^{-\beta E(\vec{r},\vec{p})}}{\partial \beta} \right) d\vec{r} d\vec{p}$$

$$U = -\frac{1}{q} \frac{\partial}{\partial \beta} \left(\frac{1}{h^3} \int \int e^{-\beta E(\vec{r},\vec{p})} \, d\vec{r} d\vec{p} \right) = -\frac{1}{q} \frac{\partial q}{\partial \beta}$$

One More Trick

$$U = -\frac{1}{q}\frac{\partial q}{\partial \beta}$$

Using $\frac{\partial(\ln y)}{\partial x} = \frac{1}{y}\frac{\partial y}{\partial x}$,

$$U = -\frac{\partial(\ln q)}{\partial \beta}$$

In general,

$$U = -\frac{\partial(\ln Q)}{\partial \beta}$$

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Entropy

- Entropy is the expectation value of $k_\mathrm{B} \ln W$

$$S = \langle k_\mathrm{B} \ln W \rangle = \frac{k_\mathrm{B}}{h^3 q} \int \int \ln W(\vec{r}, \vec{p})\, e^{-\beta E(\vec{r},\vec{p})}\; d\vec{r} d\vec{p}$$

Here,

$$W(\vec{r}, \vec{p}) = \frac{1}{P(\vec{r}, \vec{p})} = \frac{q}{e^{-\beta E(\vec{r},\vec{p})}}$$

$$S = \frac{k_\mathrm{B}}{h^3 q} \int \int \{\ln q + \beta E(\vec{r}, \vec{p})\} e^{-\beta E(\vec{r},\vec{p})}\; d\vec{r} d\vec{p}$$

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Two Terms

$$S = \frac{k_\mathrm{B}}{h^3 q} \int \int \{\ln q + \beta E(\vec{r}, \vec{p})\} e^{-\beta E(\vec{r}, \vec{p})} \, d\vec{r} d\vec{p}$$

$$S = \frac{k_\mathrm{B}}{h^3 q} \int \int \left\{ \ln q \, e^{-\beta E(\vec{r}, \vec{p})} + \beta E(\vec{r}, \vec{p}) e^{-\beta E(\vec{r}, \vec{p})} \right\} d\vec{r} d\vec{p}$$

Consider the two terms separately:

$$S = k_\mathrm{B}(I_1 + I_2)$$

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First Term

$$I_1 = \frac{1}{h^3 q} \int \int \ln q \, e^{-\beta E(\vec{r},\vec{p})} \, d\vec{r} d\vec{p}$$

$$I_1 = \frac{\ln q}{q} \times \frac{1}{h^3} \int \int e^{-\beta E(\vec{r},\vec{p})} \, d\vec{r} d\vec{p}$$

Hence,

$$I_1 = \ln q$$

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Second Term

$$I_2 = \frac{1}{h^3 q} \int \int \beta E(\vec{r}, \vec{p}) e^{-\beta E(\vec{r},\vec{p})} \, d\vec{r} d\vec{p}$$

$$I_2 = \beta \times \frac{1}{h^3 q} \int \int E(\vec{r}, \vec{p}) e^{-\beta E(\vec{r},\vec{p})} \, d\vec{r} d\vec{p} = \beta \langle E \rangle$$

Hence,

$$I_2 = \beta U$$

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Helmholtz Energy (?!)

$$S = k_{\mathrm{B}}(I_1 + I_2)$$

$$I_1 = \ln q \text{ and } I_2 = \beta U$$

Hence,

$$S = k_{\mathrm{B}} \ln q + k_{\mathrm{B}}\beta U = k_{\mathrm{B}} \ln q + \frac{U}{T}$$

Rearrangement leads to

$$-k_{\mathrm{B}}T \ln q = U - TS = A$$

In general, $A = -k_{\mathrm{B}}T \ln Q$

Partition Function and …

- Helmholtz energy: $A = -k_{\mathrm{B}}T \ln Q$
- Internal energy: $U = -\partial(\ln Q)/\partial\beta$
- Entropy: $S = (U - A)/T$

"A partition function contains

all the thermodynamic information about a system."

Derived Functions

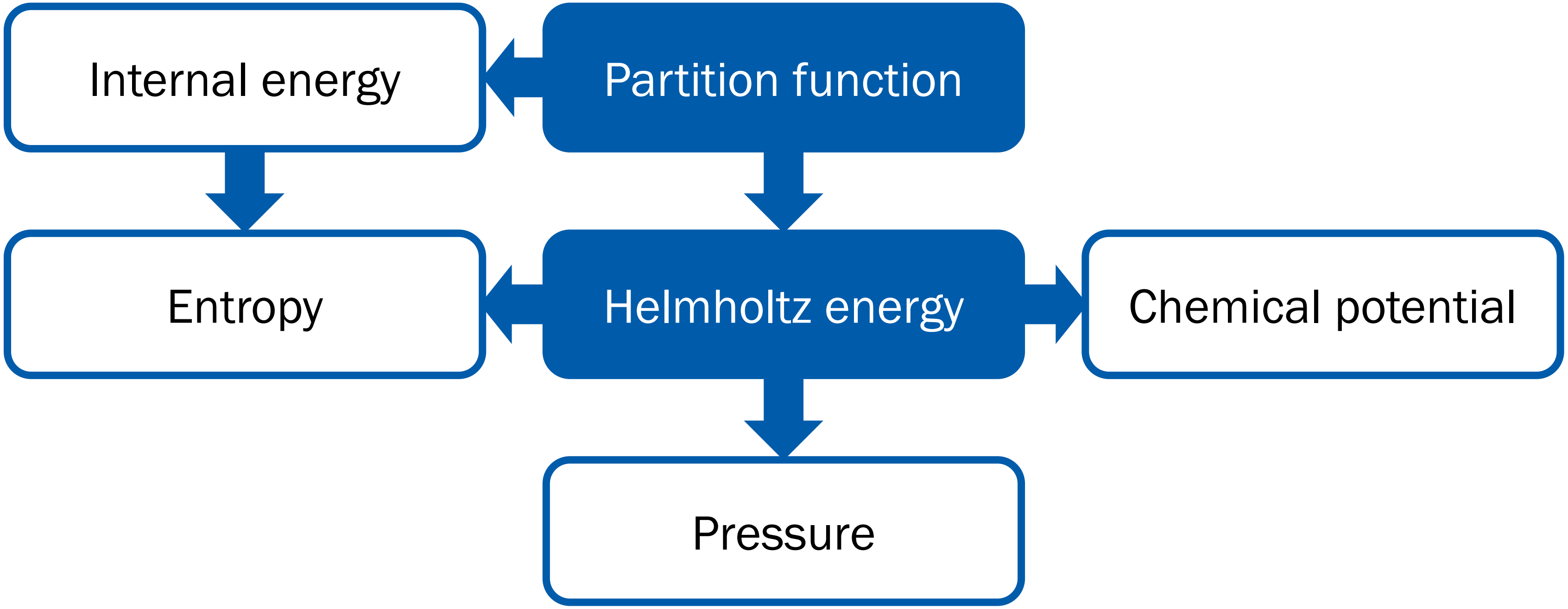
We know Helmholtz energy from the partition function,

$$A = -k_{\mathrm{B}}T \ln Q$$

Using $dA = -SdT - pdV + \mu dn$,

$$S = -\left(\frac{\partial A}{\partial T}\right)_{V,n}, \qquad p = -\left(\frac{\partial A}{\partial V}\right)_{T,n}, \qquad \mu = \left(\frac{\partial A}{\partial n}\right)_{T,V}$$

Q and Thermodynamics



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Calculation of Q

For indistinguishable particles,

$$Q = \frac{1}{N!\,h^{3N}}\left[\int\int e^{-E(\vec{r},\vec{p})/k_\mathrm{B}T}\,d\vec{r}d\vec{p}\right]^N = \frac{1}{N!}q^N$$

where

$$q = \frac{1}{h^3}\int\int e^{-E(\vec{r},\vec{p})/k_\mathrm{B}T}\,d\vec{r}d\vec{p}$$

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Perfect Gas

- No interactions (no potential energy)
- Total energy = kinetic energy $(p^2/2m)$
- Bound in a box of volume V
  - Range of x = $(0, a)$
  - Range of y = $(0, b)$
  - Range of z = $(0, c)$
  - $abc = V$
- Range of momentum = $(-\infty, \infty)$

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q of Perfect Gas

$$q = \frac{1}{h^3}\int\int e^{-E(\vec{r},\vec{p})/k_\mathrm{B}T}\, d\vec{r}d\vec{p}$$

Since $E(\vec{r},\vec{p}) = p^2/2m$,

$$q = \frac{1}{h^3}\int\int e^{-p^2/2mk_\mathrm{B}T}\, d\vec{r}d\vec{p}$$

$$q = \frac{1}{h^3}\left[\int 1\, d\vec{r}\right]\left[\int e^{-p^2/2mk_\mathrm{B}T}\, d\vec{p}\right]$$

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Position Integral

$$q = \frac{1}{h^3}\left[\int 1\, d\vec{r}\right]\left[\int e^{-p^2/2mk_\mathrm{B}T}\, d\vec{p}\right]$$

$$\int 1\, d\vec{r} = \int_0^c \int_0^b \int_0^a 1\, dx\, dy\, dz = abc = V$$

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Momentum Integral

$$q = \frac{V}{h^3}\left[\int e^{-p^2/2mk_\mathrm{B}T}\, d\vec{p}\right]$$

$$\int e^{-p^2/2mk_\mathrm{B}T}\, d\vec{p} = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-p^2/2mk_\mathrm{B}T}\, dp_x\, dp_y\, dp_z$$

Since $e^{-p^2/2mk_\mathrm{B}T} = e^{-p_x^2/2mk_\mathrm{B}T} e^{-p_y^2/2mk_\mathrm{B}T} e^{-p_z^2/m2k_\mathrm{B}T}$,

$$\left[\int_{-\infty}^{\infty} e^{-p_x^2/2mk_\mathrm{B}T}\, dp_x\right]\left[\int_{-\infty}^{\infty} e^{-p_y^2/2mk_\mathrm{B}T}\, dp_y\right]\left[\int_{-\infty}^{\infty} e^{-p_z^2/2mk_\mathrm{B}T}\, dp_z\right]$$

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Gaussian Integral Again

$$q = \frac{V}{h^3}\left[\int_{-\infty}^{\infty} e^{-p_x^2/2mk_\mathrm{B}T}\,dp_x\right]^3$$

Using $\int_{-\infty}^{\infty} e^{-\alpha x^2}\,dx = \sqrt{\pi/\alpha}$,

$$q = \frac{V}{h^3}\left[\sqrt{2\pi m k_\mathrm{B}T}\right]^3 = V\left(\frac{2\pi m k_\mathrm{B}T}{h^2}\right)^{3/2}$$

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Q of Perfect Gas

$$q = V\left(\frac{2\pi m k_{\mathrm{B}} T}{h^2}\right)^{3/2}$$

For indistinguishable gas particles,

$$Q = \frac{1}{N!}q^N = \frac{V^N}{N!}\left(\frac{2\pi m k_{\mathrm{B}} T}{h^2}\right)^{3N/2}$$

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Helmholtz Energy

$$Q = \frac{V^N}{N!}\left(\frac{2\pi m k_\mathrm{B} T}{h^2}\right)^{3N/2}$$

Now, we can calculate the Helmholtz energy:

$$A = -k_\mathrm{B} T \ln Q = -k_\mathrm{B} T \ln \left\{ \frac{V^N}{N!}\left(\frac{2\pi m k_\mathrm{B} T}{h^2}\right)^{3N/2} \right\}$$

$$A = -N k_\mathrm{B} T \ln V + k_\mathrm{B} T \ln N! - \frac{3}{2} N k_B T \ln \left(\frac{2\pi m k_\mathrm{B} T}{h^2}\right)$$

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Internal Energy

$$Q = \frac{V^N}{N!}\left(\frac{2\pi m k_\mathrm{B} T}{h^2}\right)^{3N/2}$$

Also the internal energy:

$$U = -\frac{\partial(\ln Q)}{\partial\beta} = -\frac{\partial}{\partial\beta}\ln\left\{\frac{V^N}{N!}\left(\frac{2\pi m}{\beta h^2}\right)^{3N/2}\right\}$$

$$U = \frac{\partial}{\partial\beta}\ln\beta^{3N/2} + \frac{\partial}{\partial\beta}(\text{terms independent to }\beta) = \frac{3}{2}N k_\mathrm{B} T$$

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Pressure

$$A = -Nk_{\mathrm{B}}T\ln V + k_{\mathrm{B}}T\ln N! - \frac{3}{2}Nk_BT\ln\left(\frac{2\pi mk_{\mathrm{B}}T}{h^2}\right)$$

Using $p = -(\partial A/\partial V)_{T,n}$,

$$p = Nk_BT\frac{\partial(\ln V)}{\partial V} + \frac{\partial}{\partial V}(\text{terms independent to } V)$$

$$p = \frac{Nk_{\mathrm{B}}T}{V}$$

One can obtain entropy and chemical potential similarly

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Real Gas

Particles are no more independent to each other

$$Q \neq q^N / N!$$

$$Q = \frac{1}{N!\,h^{3N}} \int \cdots \int e^{-E_{\mathrm{tot}}/k_{\mathrm{B}}T}\, d\vec{r}_1 d\vec{p}_1 d\vec{r}_2 d\vec{p}_2 \cdots d\vec{r}_N d\vec{p}_N$$

where

$$E_{\mathrm{tot}} = \sum_i \frac{p_i^2}{2m} + U(\vec{r}_1, \vec{r}_2, \cdots, \vec{r}_N)$$

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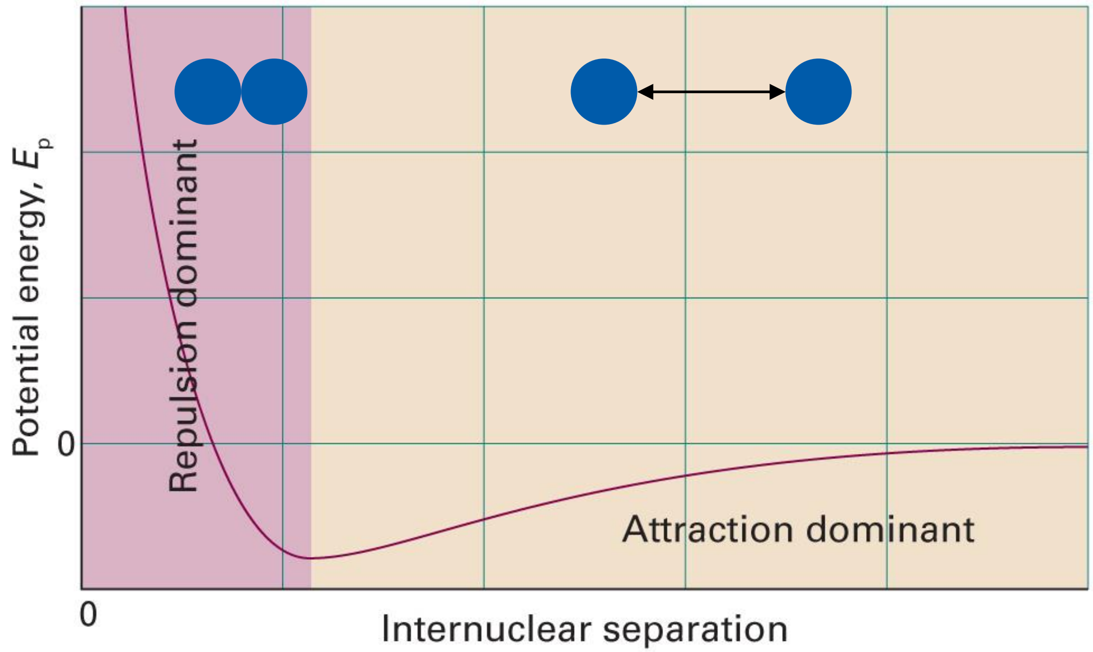
Potential Energy

Pairwise interactions (attraction + repulsion)



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Figure 1C.1)

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Pressure of Real Gas

Partition function

↓

Helmholtz energy

↓

Pressure

$$p = \frac{nRT}{V - nb} - a\frac{n^2}{V^2}$$

Equipartition Theorem

For a sample at thermal equilibrium,

the average value of each quadratic contribution

to the internal energy is $(1/2)k_{\mathrm{B}}T$.

(Quadratic contribution: e.g., $E \propto p_x^2$ or $E \propto x^2$)

Can we prove this theorem?

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Setting up the Stage

$$Q = \frac{1}{N!\,h^{3N}} \int \cdots \int e^{-E_{\text{tot}}/k_{\text{B}}T}\, d\vec{r}_1 d\vec{p}_1 d\vec{r}_2 d\vec{p}_2 \cdots d\vec{r}_N d\vec{p}_N$$

Assume that

$$E_{\text{tot}} = \sum_{i=1}^{f_r} A_i r_i^2 + \sum_{j=1}^{f_p} B_j p_j^2$$

$$(A_i, B_j > 0)$$

and that the ranges of both positions and momenta are $(-\infty, \infty)$

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Summation to Product

$$Q = \frac{1}{N!\,h^{3N}} \int \cdots \int e^{-\left(\sum_{i=1}^{f_r} A_i r_i^2 + \sum_{j=1}^{f_p} B_j p_j^2\right)/k_\mathrm{B}T}\, d\vec{r}_1 d\vec{p}_1 d\vec{r}_2 d\vec{p}_2 \cdots d\vec{r}_N d\vec{p}_N$$

$$= \frac{1}{N!\,h^{3N}} \int \cdots \int \prod_{i=1}^{f_r} e^{-A_i r_i^2/k_\mathrm{B}T} \prod_{j=1}^{f_p} e^{-B_j p_j^2/k_\mathrm{B}T}\, d\vec{r}_1 d\vec{p}_1 d\vec{r}_2 d\vec{p}_2 \cdots d\vec{r}_N d\vec{p}_N$$

Thus,

$$Q = \frac{1}{N!\,h^{3N}} \prod_{i=1}^{f_r} \int_{-\infty}^{\infty} e^{-A_i r_i^2/k_\mathrm{B}T}\, dr_i \prod_{j=1}^{f_p} \int_{-\infty}^{\infty} e^{-B_j p_j^2/k_\mathrm{B}T}\, dp_j$$

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Gaussian Integrals

$$Q = \frac{1}{N!\, h^{3N}} \prod_{i=1}^{f_r} \int_{-\infty}^{\infty} e^{-A_i r_i^2/k_\mathrm{B}T}\, dr_i \prod_{j=1}^{f_p} \int_{-\infty}^{\infty} e^{-B_j p_j^2/k_\mathrm{B}T}\, dp_j$$

Using

$$\int_{-\infty}^{\infty} e^{-A_i r_i^2/k_\mathrm{B}T}\, dr_i = \sqrt{\frac{\pi k_\mathrm{B} T}{A_i}}, \qquad \int_{-\infty}^{\infty} e^{-B_j p_j^2/k_\mathrm{B}T}\, dp_j = \sqrt{\frac{\pi k_\mathrm{B} T}{B_j}},$$

we obtain

$$Q = \frac{1}{N!\, h^{3N}} \prod_{i=1}^{f_r} \left(\frac{\pi k_\mathrm{B} T}{A_i}\right)^{1/2} \prod_{j=1}^{f_p} \left(\frac{\pi k_\mathrm{B} T}{B_j}\right)^{1/2}$$

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In Terms of $\boldsymbol{\beta}$

$$Q = \frac{1}{N!\,h^{3N}} \prod_{i=1}^{f_r} \left(\frac{\pi k_{\mathrm{B}} T}{A_i}\right)^{1/2} \prod_{j=1}^{f_p} \left(\frac{\pi k_{\mathrm{B}} T}{B_j}\right)^{1/2}$$

We can write the partition function as follows:

$$Q = \frac{\pi^{(f_r+f_p)/2}}{N!\,h^{3N}} \left(\frac{1}{\beta}\right)^{(f_r+f_p)/2} \prod_{i=1}^{f_r} \left(\frac{1}{A_i}\right)^{1/2} \prod_{j=1}^{f_p} \left(\frac{1}{B_j}\right)^{1/2}$$

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Internal Energy

$$Q = \frac{\pi^{(f_r+f_p)/2}}{N!\,h^{3N}}\left(\frac{1}{\beta}\right)^{(f_r+f_p)/2}\prod_{i=1}^{f_r}\left(\frac{1}{A_i}\right)^{1/2}\prod_{j=1}^{f_p}\left(\frac{1}{B_j}\right)^{1/2}$$

The internal energy can be calculated from

$$U = -\left(\frac{\partial \ln Q}{\partial \beta}\right)$$

Now,

$$\ln Q = -\frac{f_r + f_p}{2}\ln\beta + (\text{terms independent to } \beta)$$

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Internal Energy

$$\ln Q = -\frac{f_r + f_p}{2} \ln \beta + (\text{terms independent to } \beta)$$

$$U = -\left(\frac{\partial \ln Q}{\partial \beta}\right) = \frac{f_r + f_p}{2} \frac{\partial \ln \beta}{\partial \beta} = \frac{f_r + f_p}{2} \frac{1}{\beta} = \frac{f_r + f_p}{2} k_{\mathrm{B}} T$$

Therefore,

$$U = \left(f_r + f_p\right) \frac{k_{\mathrm{B}} T}{2}$$

"The average value of each quadratic contribution

to the internal energy is $(1/2)k_{\mathrm{B}}T$."

Summary

$$S = k_{\mathrm{B}} \ln W$$

Boltzmann factor

$$P(s) = \frac{1}{q} e^{-E(s)/k_{\mathrm{B}}T}$$

- Partition function

$$q = \sum e^{-E(s)/k_{\mathrm{B}}T}$$

Classical Stat Mech

Phase space

Partition function

$$q = \frac{1}{h^3} \iint e^{-E/k_{\mathrm{B}}T}\, d\vec{r}\, d\vec{p}$$

- Many-particle PF

$$Q = \frac{1}{N!} q^N$$

Thermodynamics

- Helmholtz energy
- Internal energy

⋮

Perfect gas

- Partition function
- Helmholtz energy
- Pressure

Perfect gas law

Real gas

Equipartition theorem

- Partition function
- Internal energy